Nila Mönig, PhD

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PROFILE

PhD neuroscientist with 10+ years in connectomics and electron microscopy. Develops deep-learning pipelines for large, complex image data, together with the tools, quality control, documentation and partnerships that make them usable. Hands-on from hardware to code: builds solutions where no standard tool exists.

CORE STRENGTHS

- **Computational imaging and machine learning:** deep-learning segmentation and classification of TB-scale electron-microscopy data, deployed in parallel on HPC.
- **Bridging disciplines and partners:** 10+ years in interdisciplinary teams across biology, computation, imaging and external partners.
- **Building tools and hardware:** custom proofreading tools (napari, Neuroglancer) and self-built behavioural hardware.
- **Communication and knowledge transfer:** advises colleagues on connectomics methods; thesis, two publications, two preprints, six posters, talks, tool documentation and an annotation guideline.

CORE COMPETENCES

Computational microscopy: Electron microscopy | Image analysis | Computer vision | Segmentation
Machine learning & AI: PyTorch | MONAI | Gunpowder | scikit-learn | Weights & Biases | U-Net / ResNet | Object classification | Ground-truth generation | Model evaluation
Programming & tools: Python | MATLAB | Git | napari | Neuroglancer | BigWarp | FIJI/ImageJ | ilastik
Workflow & reproducibility: Automation | Versioned pipelines | Documentation | Quality control | Benchmarking
Scalable data processing: SLURM/HPC | Array-based deployment | Parallel inference | Process automation
Data analytics: Statistical analysis | Multi-dimensional data analysis | Clustering | Dimensionality reduction
Hardware and prototyping: Prototyping | Fabrication | Fluidics | Control electronics |Soldering
Collaboration & communication: Cross-disciplinary collaboration | Partner coordination | Mentoring and knowledge transfer | Scientific writing | Presentation | Scientific figure and poster design (Adobe Illustrator)
Experimental methods (earlier in career): Behavioural experiments in adult zebrafish | Electrophysiology | Pharmacology | Immunohistochemistry | Neuron tracing | Cell culture

PROFESSIONAL EXPERIENCE

Independent Work
Jul 2026 – present

- Finalising the synapse-detection pipeline (code cleanup, documentation, validation) for release
- Revising a manuscript for journal publication

POSTDOCTORAL SCIENTIST
Friedrich Miescher Institute for Biomedical Research, Basel – Oct 2022 – Jun 2026

Led development of scalable data pipelines for connectomics analysis in zebrafish neuroscience

- **Designed and implemented**, as sole developer, a high-throughput synapse-detection pipeline tailored to non-standard tissue properties.
- **Developed, trained, and deployed** deep-learning segmentation and classification models (U-Net, ResNet variants) on HPC infrastructure using array-based parallel deployment.
- **Tool development:** created and maintained custom proofreading and training-data tools in napari and Neuroglancer, with documentation and user feedback integrated.
- **Partner coordination:** set up and coordinated a high-throughput proofreading partnership with an external company (ground-truth generation, neuron reconstruction at scale); established benchmarking protocols; ran quality assessment and error feedback (incl. review meetings) for ground truth; wrote interface scripts to reconcile platforms and data formats for the reconstructions, and reconstructed a subset of neurons.

- **Segmentation and ground truth:** took over responsibility, on our side of a collaboration with Google, for the neuron segmentation of a large, multimodal dataset (EM, calcium imaging, behavioural training) with highly variable image quality; with models developed by the partner, inspected output, decided on its adequacy, masked data, generated and corrected ground truth; contributed to mapping imaged cells onto the EM volume (BigWarp).
- **Strategy and data quality:** designed the experimental workflow in own projects and the computational strategy of the synapse-detection pipeline; gave image-quality feedback to acquisition teams to support segmentation readiness across datasets of different origins.
- **Knowledge transfer:** advised colleagues on segmenting their own datasets (timelines, steps from raw data to segmentation); introduced connectomics methods; accompanied them in their interactions with the proofreading company; supervised an intern in annotation and tool use.

PHD CANDIDATE
Friedrich Miescher Institute for Biomedical Research – May 2012 – Sep 2022
Thesis: Neuronal Organisation of the Olfactory Bulb in Adult Zebrafish | magna cum laude (5.5/6.0)

Developed integrated connectomics and behavioural-analysis workflows to investigate neural-circuit function in the zebrafish olfactory system.

- **Collaboration with Google Research:** partnered on neuron segmentation for TB-scale EM data; assessed model output by visual inspection, reported errors and generated targeted ground-truth data to close gaps.
- **Proofreading tools:** developed a Neuroglancer-based interface for inspecting and curating automated segmentations (15–50× faster reconstruction) and a tool for manual neuron classification.
- **Analysis:** proofread >450 neurons and established a morphological classification scheme.
- **Process automation:** co-developed and automated behavioural assays for odour discrimination in adult zebrafish (87% less manual work, 2x throughput); implemented a multi-dimensional analysis pipeline.

RESEARCH ASSISTANT, University of Freiburg – Jan 2011 – Mar 2012
- **Prepared and maintained** primary neuronal cell cultures ensuring viability and experimental quality.
- **Implemented** closed-loop stimulation circuits (Tucker Davis Technologies); performed electrophysiology, pharmacology, immunohistochemistry and neuron tracing.

STUDENT ASSISTANT, University of Freiburg – Jun 2007 – Sep 2009
- **Teaching assistant** for introductory MATLAB course.
- **Contributed** to data-management system development and neuroanatomical analyses.

EDUCATION
- PhD Neuroscience, University of Basel (2022) – magna cum laude
- Diplom Biology (≈ MSc), University of Freiburg (2009) – 1.0 (best possible)

SELECTED PUBLICATIONS & PRESENTATIONS
- Mönig N.R.*, Januszewski M.* et al., 2026, **eLife** Reviewed Preprint 112715
- Montaño Crespo R.E., Mönig N.R. et al., 2026, **bioRxiv** 2026.09.18.752720 (preprint)
- Frank T., Mönig N.R., Satou C. et al., **Nat Neurosci** 22: 1844–1856 (2019)
- Namekawa I.*, Mönig N.R.*, Friedrich R.W. **Exp Brain Res** 236: 2959–2969 (2018)

(**equal contribution*)
Posters: Connectomics (2019, 2022, 2024), ECRO (2022), Zebrafish Brain Conference (2022), FENS (2018). Talks at regional meetings.

SELECTED COURSES & TRAINING
- Entrepreneurship Training, Innosuisse Module 2 (Innovation Office, University of Basel, 2026)
- Advanced Python and Machine Learning (University of Basel, 2024)
- Deep Learning for image analysis (EMBL, Heidelberg, 2023)

LANGUAGES
German (native), English (C2), French (B1)

INTERESTS
Volunteer conservation work, running, healthy eating and cooking, furniture upcycling, knitting